

Anomalous Magnetic Moments in FFGFT Geometric Derivation from the Time–Mass Duality

Purely Geometric Formulas and Precise Ratio Predictions

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Ideal anchor number: $f = 1/(4\xi) \rightarrow f = 1/\xi = 30000/4 = 7500$ ($1/(4\xi)$ would give 1875).
- Number of divisors of $7500 = 2^2 \cdot 3 \cdot 5^4$: $36 \rightarrow 30$.
- Perspective 2 for G : $G = \frac{\xi}{2} \cdot 10^{-6}$: $6.674 \rightarrow 6.667 \times 10^{-11}$, deviation $0.01\% \rightarrow -0.11\%$ (also in the comparison list and the table).
- Quark formulas with $f = 7500$: m_u 2.3 $\rightarrow$ 60.5 MeV, m_d 4.8 $\rightarrow$ 80.6 MeV (factor 28 and 17 above the measured value), m_s 95 $\rightarrow$ 156 MeV (+67%), m_c 1.27 $\rightarrow$ 2.73 GeV (+115%), m_b 4.2 $\rightarrow$ 4.42 GeV (+5.7%); no common anchor explains the deviations. m_t : deviation $0.87\% \rightarrow 1.0\%$.
- $f^4 = 7491.91^4 = 3.150 \times 10^{15}$ instead of 3.155×10^{15} (already corrected in the text on 29 Sept. 2026). On the formula for v as $v/m_P = \xi^4/(5\pi)$ see Doc. 383.
- Dark energy (1 Oct. 2026): ρ_Λ $7.96 \times 10^{-27} \rightarrow 7.85 \times 10^{-26}$ kg/m³ (factor 13 above the measured value).
- Calculation check, further points: $k_{\text{geom}}^{\text{rec}}$: $2.272 \rightarrow 1.446$ (with $k_{\text{geom}} = 2.2236$, a_e is 35 % too small, the "approximately 2 %" do not hold); deviations m_e : $1.02 \rightarrow -0.88\%$, m_τ : $2.0 \rightarrow +2.2\%$; unit of $c^5/\hbar$: $\text{m}^3/(\text{kg}\cdot\text{s}) \rightarrow \text{m}^3/(\text{kg}\cdot\text{s}^4)$; $2\pi^3$ is not a doubled sphere volume; m_Z/m_W : scheme mix-up, no 0.5 % deviation; coupling ratio "about 1/10" $\rightarrow \sin \theta_W \approx 0.48$; notes on the calculations with $f = 7491.91$ instead of 7500, on 5φ (0.108 % instead of $\sim 0.5\%$) and on $f^{1/3}$ versus $f^{1/3} - 1$.
- (2 Oct. 2026) Factor 10 in v : fitted to the measured value, not motivated; with the bare v of the ladder $10 K_{\text{frak}} \approx \pi^2$ (Doc. 387).
- (2 Oct. 2026) Notation f : base winding number $f = 1/\xi$ — not a frequency; note in the text (Doc. 190, R147).

(Note of 2 Oct. 2026 on notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{T0}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135). Cf. Doc. 190, R147.)

Abstract

In the present work, the fundamental architecture of spacetime is reinterpreted within the framework of the **Fundamental Fractal Geometric Field Theory (FFGFT)** – internally referred to as the T0 model. The central paradigm consists in the transition from a point-like to a purely geometric description of the vacuum as a four-dimensional **gyral torus**.

Geometric structure: The theory is founded on the fractal-geometric basic structure with the parameter $\xi \approx (4/3) \times 10^{-4}$ and the densest local sphere packing through regular **tetrahedra**. This tetrahedral basis forms the stable foundation for the low generations (electron, muon, proton/neutron) as well as the local 3D crystal structure of the torso. Building upon this, through fractal branching and pentagonal symmetry breaking, the ideal sub-Planck factor

$$f = 7500,$$

emerges, which represents an exactly 7500-fold reduction compared to the conventional Planck scale (t_0) and follows directly from the geometric winding density $30000/4$.

g-2 anomaly: A centerpiece of this work is the transparent geometric derivation of the anomalous magnetic moments of leptons. While the Standard Model relies on numerous perturbative terms, in the FFGFT the electron anomaly follows directly from the base winding (tetrahedral projection). The muon and tau anomalies arise through fractal branchings with Hausdorff dimensions $p \approx 5/3$ and $4/3$, respectively. With the ideal value $f = 7500$, the purely geometric predictions achieve an accuracy of approximately 2 %. (*Note of 30 Sept. 2026: this does not hold for a_e – with $k_{geom} = 2.2236$, $a_e = (4\pi/f)/k_{geom} = 7.54 \times 10^{-4}$, –35 % relative to 1.15965×10^{-3} ; see calculation example 3.) By reconstructing the projection factor k_{geom} , the deviation for the muon drops to below 0.2 %. The most precise, k_{geom} -independent prediction for the tau anomaly reads*

$$a_\tau \approx 1.282 \times 10^{-3},$$

which follows exclusively from the exact ratio $f^{1/3} - 1$.

Geometric proportionality: All basic physical quantities (constants, masses, couplings) are traced back to fixed geometric ratios; the theory works with the single parameter ξ (plus motivated integer structural choices), while SI anchors serve for unit conversion. T0 theory thus offers a transparent geometric description and delivers concrete, experimentally verifiable predictions – in particular for the tau anomaly as a clear test at Belle II.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction: The Geometric Paradigm

Open Questions of Modern Physics

The 21st century faces a basic question: While the Standard Model of particle physics describes experimental data with very high precision, it nevertheless contains ~19 free parameters that cannot be derived from principles but must be empirically adjusted. In addition, this model predicts no values for fundamental constants such as the fine-structure constant α , the masses of the electron or proton, or the strength of gravity.

At the same time, there are hints of phenomena that may point beyond the Standard Model: The observed acceleration of cosmic expansion (dark energy), the anomalies in galaxy rotation curves (dark matter), and the precise measurements of anomalous magnetic moments of leptons all show discrepancies with established theory.

T0 theory takes a different approach: Instead of postulating new particles or fields, it starts from a fundamental geometric structure of spacetime itself.

The Basic Idea: Spacetime as a Torsion Crystal

The central thesis of T0 theory can be summarized in one sentence:

The universe is a static 4-dimensional torsion crystal whose discrete sub-Planck structure generates all observable physical phenomena.

What does this mean concretely?

1. **Static [S]:** The universe does not expand in the conventional sense. The observed redshift arises from geometric path elongation in the torsion lattice. (Cosmological statement; the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either, P39.)

2. **4-dimensional:** Besides the three spatial dimensions, a fourth exists that is not identical to time but represents an additional spatial dimension that is "rolled up" in our experiential space.
3. **Torsion crystal:** Spacetime is not continuous but possesses a discrete, crystalline structure at the sub-Planck scale. The "torsion" describes the windings and twists of this crystal structure.
4. **Sub-Planck structure:** The fundamental length scale is not the Planck length $\ell_p = 1.616 \times 10^{-35}$ m, but a scale smaller by the factor $f = 7491.91$. (Note of 30 Sept. 2026: the document sets $f = 1/\xi = 7500$ without subtracting 5φ (sections on the anchor number and symmetry breaking); $7491.91 = 7500 - 5\varphi$. In the calculations with 7491.91, using $f = 7500$ changes: $v = 245.65$ GeV, $m_e = 0.5038$ MeV (−1.4 %), $m_\mu = 102.90$ MeV (−2.6 %), $f^{1/3} = 19.574$.)

In this picture, **particles are not point-like objects** but standing waves (resonances) in the torsion crystal. **Forces** are not exchange of virtual particles but geometric couplings between different torsion modes. **Masses** are not intrinsic properties but frequencies of these resonances.

.2 The Fundamental Derivation: From Geometry to Numerical Values

The Narrative Starting Point: Why 30000?

The derivation begins with a seemingly arbitrary number: 30000. It is a motivated setting: it is meant to encode the structure of 4-dimensional spacetime. Imagine: We live in a world with **three** experienceable spatial dimensions. But at the most fundamental level, a **fourth** dimension exists that is not directly accessible but only indirectly perceptible through its geometric effects. This fourth dimension is "compactified" – it is rolled up at the smallest scales. The number 30000 arises from the interaction between these four dimensions:

- The **3** stands for the three experienceable spatial dimensions.
- The **4** stands for the full, four-dimensional reality.
- The **000** (i.e., factor 1000) describes the scale hierarchy between the fundamental and the observable level.

Concretely, we define:

$$\xi = \frac{4}{30000} = 1.333\bar{3} \times 10^{-4} \quad (1)$$

This number ξ is the **fundamental correction parameter**. It describes how strongly the real 4D spacetime deviates from an ideal 3D geometry. Physically interpreted: ξ is the "torsion strain" – the tiny twist that distinguishes the spacetime lattice from a perfect structure.

The Ideal Anchor Number: Why 7500?

From ξ , the ideal anchor number is obtained:

$$f = \frac{1}{\xi} = \frac{30000}{4} = 7500 \quad (2)$$

(Corrected on 30 Sept. 2026: previously $f = 1/(4\xi)$ – with $\xi = 4/30000$, $1/(4\xi) = 1875$; $30000/4 = 7500$ is $1/\xi$.) This is the number referred to as the ideal sub-Planck factor: The

ideal anchor number of the crystal lattice. **Why is 7500 so special?** Let us look at its prime factorization:

$$7500 = 2^2 \times 3 \times 5^4 = 4 \times 3 \times 625 \quad (3)$$

The number has a rich divisor structure:

- It has **30 positive divisors** – favourable for a symmetric lattice structure. (*Corrected on 30 Sept. 2026: previously $36 = 2^2 \cdot 3 \cdot 5^4$ has $3 \cdot 2 \cdot 5 = 30$ divisors.*)
- It combines the first three prime numbers (2, 3, 5).
- The factor $5^4 = 625$ points to the pentagonal symmetry of the crystal (5) in four dimensions (exponent 4).
- The number is divisible by numerous factors – a favourable basis for resonances.

In crystallography, structures with many divisors are called "highly symmetric" – which fits a fundamental spacetime structure.

The Symmetry Breaking: The Role of the Golden Ratio

A perfect, ideal crystal would be completely symmetric. But our world shows symmetry breaking at all levels:

- Matter dominates over antimatter
- The weak interaction violates parity symmetry
- The neutron is heavier than the proton
- The three generations of leptons have different masses

T0 theory proposes **[S]** a common geometric origin for these symmetry breakings: the pentagonal symmetry of the crystal, embodied by the **golden ratio** φ . The golden ratio $\varphi = (1 + \sqrt{5})/2 = 1.618033989 \dots$ is the irrational number that describes pentagonal symmetry. In a perfect pentagon, φ appears everywhere: The ratio of diagonal to side is exactly φ . Why pentagonal symmetry specifically? For mathematical reasons (crystallographic restriction), pentagonal symmetry is the first that **cannot tile periodically** in the plane. This leads to "quasicrystals" – structures that are ordered but not periodic. Exactly such a quasicrystalline structure is postulated by T0 theory for the sub-Planck scale. The symmetry breaking is not quantified in the theory by a direct subtraction of 5φ from the ideal anchor number 7500. Instead, it is hidden in the **approximately 2 % deviations** that appear in the calculations of anomalous magnetic moments (g-2 anomalies). This deviation arises from the pentagonal projection in the geometric factor k_{geom} :

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \approx 2.22357, \quad (4)$$

which projects the 4D torsion onto the 3D world. The reconstructed version from experimental data deviates by about 2 % ($k_{\text{geom}}^{\text{rec}} \approx 2.26955$) (*Note of 30 Sept. 2026: reconstructed from a_e^{exp} , $k_{\text{geom}}^{\text{rec}} = (4\pi/f)/a_e^{\text{exp}} = 1.446$, not ≈ 2.27 ; see calculation example 3*), which reflects the actual symmetry breaking – a slight distortion through the pentagonal geometry that breaks perfect symmetry without changing the ideal value $f = 7500$.

The Real Sub-Planck Factor: $f = 7500$

Now we put everything together: The ideal crystal remains intact, the symmetry breaking only affects the projection factors:

$$\boxed{f = 7500} \quad (5)$$

This is a **central number of T0 theory**. It appears in almost all formulas and describes:

- The number of sub-Planck cells per Planck length
- The density of the torsion lattice
- The fundamental frequency of all geometric resonances

.3 Stage 1: From Geometry to Energy – the Higgs Field

The Planck Scale as Natural Reference

In theoretical physics, there exists a natural scale for mass, length, and time: the Planck scale. It results from a combination of fundamental constants:

$$m_P = \sqrt{\frac{\hbar c}{G}} = 1.220910 \times 10^{19} \text{ GeV}/c^2 \quad (6)$$

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} = 1.616255 \times 10^{-35} \text{ m} \quad (7)$$

$$t_P = \sqrt{\frac{\hbar G}{c^5}} = 5.391247 \times 10^{-44} \text{ s} \quad (8)$$

These quantities mark the scale at which quantum effects of gravity become important. In conventional physics, it remains unclear why observable particle masses are so much smaller than the Planck mass (the hierarchy problem).

In T0 theory, the Planck scale receives a geometric interpretation: It is the **lattice vibration frequency** of the fundamental crystal. The Planck mass is the energy required to maximally excite a single lattice cell.

The 4D Energy Density: Dilution over Four Dimensions

The basic assumption of this section is: The Planck energy is not concentrated on a single cell but distributed over the four-dimensional lattice. Why four dimensions? Because each of the four spatial dimensions of the torsion crystal contributes to the energy density.

Mathematically, this means:

$$\rho_{4D} = \frac{m_{\text{Planck}}}{f^4} \quad (9)$$

Narrative explanation: Imagine a perfect cube whose edge length is f cells. In three dimensions, this cube contains f^3 cells. In four dimensions, the hypercube contains f^4 cells. The Planck energy, originally concentrated on a single cell, is now distributed uniformly over all f^4 cells of the four-dimensional hypercube.

Let us calculate:

$$f^4 = 7491.91^4 \approx 3.150 \times 10^{15} \quad (10)$$

The 4D energy density is thus smaller than the Planck mass by a factor of 3.150×10^{15} :

$$\rho_{4D} = \frac{1.220910 \times 10^{19} \text{ GeV}}{3.150 \times 10^{15}} \approx 3.875 \times 10^3 \text{ GeV} \quad (11)$$

(Corrected on 29 Sept. 2026: previously 3.155×10^{15} and 3869 GeV – $7491.91^4 = 3.150 \times 10^{15}$, as inserted correctly further below for v .)

We obtain an energy density of approximately 3875 GeV. This is still much higher than observable energy scales, but we are on the right track.

Projection onto 3D: The Half-Space Effect

We live in a three-dimensional world. The fourth dimension is not directly accessible to us. How does the energy density from the fourth dimension reach our three-dimensional experiential world?

This occurs through **geometric projection**. Imagine a 4D sphere (a 3-sphere) being projected into our 3D world. The projection of a full 4D sphere onto the 3D half-space occurs through division by $\pi/2$.

Why exactly $\pi/2$? Consider the simpler 2D case: The projection of a semicircle (angle π) onto a line yields a factor of $\pi/2$. Analogously, the projection of a 3-sphere (surface area: $2\pi^2$) onto the 3D half-space is given by $\pi/2$.

Scaling to the Electroweak Scale: The Factor 1/10

The 3D energy density obtained after projection must still be scaled to the electroweak scale. The transition from the fundamental geometric scale to the electroweak scale requires a further scaling by factor 1/10.

Why 1/10? The factor is a motivated setting; there are several interpretations of it:

1. It describes the effective dimension of electroweak theory.
2. It is associated with the ratio of electric to weak coupling; this ratio is, however, $e/g = \sin \theta_W \approx 0.48$ (or $\alpha/\alpha_W = \sin^2 \theta_W \approx 0.23$), not about 1/10.
3. It is close to the square root of the fine-structure constant ($\sqrt{\alpha} \approx 0.085$).

(Note of 2 Oct. 2026: the factor 10 is neither derived nor motivated but fitted to the measured value: in the working state of February 2026, $m_P/(f^4 \cdot \pi/2)$ gave 2467 GeV instead of 246 GeV, and the gap was closed with 1/10. If the bare $v = 248.93$ GeV of the ladder is inserted instead of the measured v , the formula requires $10 K_{\text{frak}} \approx \pi^2$ instead of 10; the factor 10 would then be a mix-up of the bare ladder value with the SI comparison value (choice, open). Comparison of the variants in Doc. 387.)

The Final Result: The Higgs VEV

In summary, we obtain:

$$v = \frac{m_P}{f^4 \cdot (\pi/2) \cdot 10} \quad (12)$$

Inserting numerical values:

$$f^4 = 7491.91^4 = 3.150 \times 10^{15} \quad (13)$$

$$v = \frac{1.220910 \times 10^{19}}{3.150 \times 10^{15} \cdot (\pi/2) \cdot 10} \quad (14)$$

$$= 246.71 \text{ GeV} \quad (15)$$

Experimental value: $v_{\text{exp}} = 246.22 \text{ GeV}$

Precision:

$$\frac{|246.71 - 246.22|}{246.22} = 0.00199 = 0.20\% \quad (16)$$

(Added on 29 Sept. 2026: with $f = 1/\xi = 7500$ the same formula gives $v = 245.65 \text{ GeV}$ (-0.23%). Since $f = 1/\xi$, it reads $v/m_P = \xi^4/(5\pi)$; this allows v to be eliminated from the lepton ladder, so that masses, Planck scale and G hang on a single anchor. See Doc. 383.)

From geometric principles – the four-dimensional dilution of Planck energy, the projection onto 3D, and the scaling to the electroweak scale – the Higgs vacuum expectation value is obtained with the deviation given above.

.4 The Fine-Structure Constant α : Two Complementary Approaches

The fine-structure constant $\alpha \approx 1/137$ describes the strength of the electromagnetic interaction. In contrast to standard physics, which regards α as a purely empirical value, the T0 model offers two independent theoretical approaches: a time-based (geometric) and an energy-based path.

The Time-Based Path (Geometric)

The first derivation considers α^{-1} as a projection of a 4D torsion wave into 3D space:

$$\alpha^{-1} = (f_{\text{ideal}} \cdot \xi) \cdot \pi^4 \cdot \sqrt{2} \quad (17)$$

Since $f_{\text{ideal}} \cdot \xi = 7500 \cdot (4/30000) = 1.0$ **exactly**, this simplifies to:

$$\alpha^{-1} = \pi^4 \cdot \sqrt{2} = 97.409 \cdot 1.414 = 137.757 \quad (18)$$

Detailed calculation:

$$\pi^4 = 97.409091 \quad (19)$$

$$\sqrt{2} = 1.414214 \quad (20)$$

$$\pi^4 \cdot \sqrt{2} = 137.757258 \quad (21)$$

Interpretation: In this derivation, the fine-structure constant appears as a **geometric number** (with the deviation given below). It follows from π (circle) and $\sqrt{2}$ (square diagonal). The lattice unit $f_{\text{ideal}} \cdot \xi = 1$ normalizes the electromagnetic coupling strength to the fundamental unit of the torsion lattice.

CODATA reference value: $\alpha_{\text{exp}}^{-1} = 137.035999177$

Deviation from the CODATA value:

$$\frac{|137.757 - 137.036|}{137.036} = 0.00526 = 0.526\% \quad (22)$$

The Energy-Based Path (Field Coupling)

The second approach uses a characteristic energy scale E_0 :

$$\alpha = \xi \cdot E_0^2 \quad (23)$$

The energy scale E_0 is meant to arise from the lattice structure; here it is determined from the measured value:

$$E_0 = \sqrt{\frac{\alpha_{\text{exp}}}{\xi}} = \sqrt{\frac{1/137.036}{1.333 \times 10^{-4}}} \approx 7.398 \text{ MeV} \quad (24)$$

Thus:

$$\alpha = \xi \cdot E_0^2 = \frac{4}{30000} \cdot (7.398)^2 = \frac{4 \cdot 54.73}{30000} = \frac{218.9}{30000} = \frac{1}{137.04} \quad (25)$$

Deviation from the CODATA value:

$$\frac{|137.04 - 137.036|}{137.036} = 0.00003 = 0.003\% \quad (26)$$

Interpretation: This approach shows α as a function of a characteristic energy scale $E_0 \approx 7.4 \text{ MeV}$. **[S]** Since E_0 is calculated here from α_{exp} , the agreement (0.003%) is a consistency check and not yet an independent prediction; an independent derivation of E_0 from the lattice structure is still missing in this document.

Comparison and Interpretation

Method	α^{-1}	Deviation
CODATA (experimental)	137.035999	Reference
Time-based (geometric)	137.757	+0.526%
Energy-based (field coupling)	137.04	+0.003%

Table 1: Comparison of the two theoretical T0 approaches with the experimental value.

The $\sim 0.5\%$ difference between the two approaches is interpreted here as the expression of two different physical aspects:

- **Time-based (geometric):** Describes the ideal lattice without dynamic effects. Shows the pure geometric structure from π and $\sqrt{2}$.
- **Energy-based (field coupling):** Is meant to describe the real field coupling with vacuum polarization and other quantum effects. Agreement 0.003% (by construction, see above).

The difference of $\sim 0.5\%$ corresponds to the pentagonal symmetry breaking $\Delta = 5\varphi$, which also appears in $f = f_{\text{ideal}} - \Delta$. If this assignment holds **[S]**, it would support the internal consistency of T0 theory: the same geometric symmetry breaking would show up in several fundamental constants. (*Note of 30 Sept. 2026: $5\varphi/7500 = 0.108\%$, not $\sim 0.5\%$; even with $f = 7491.91$ the time-based approach keeps $\alpha^{-1} = \pi^4\sqrt{2} = 137.76$ (+0.526%), so 5φ does not explain the difference.*)

Key statement: Both approaches are understood here as complementary. The time-based approach shows the ideal geometry, the energy-based one the real field coupling. Together they give a coherent picture of the fine-structure constant.

.5 The Gravitational Constant: Three Perspectives on ONE Constant

Important preliminary remark: The following three formulas describe **not** three different gravitational constants, but **a single** constant G from three mathematically equivalent perspectives.

The gravitational constant $G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ describes the strength of gravity. Compared to the electromagnetic force, it is about 10^{36} weaker. In T0 theory, this extreme weakness is traced back to the geometric structure of spacetime.

Perspective 1: Time Structure (Micro Level)

The first perspective derives G from the fundamental sub-Planck time scale:

$$G = (t_0 \cdot f)^2 \cdot \frac{c^5}{\hbar} \quad (27)$$

Geometric component: $(t_0 \cdot f)^2$ [Dimension: s^2]

SI conversion: $c^5/\hbar$ [unit conversion only]

Calculation:

$$t_0 = 7.188310237 \times 10^{-48} \text{ s} \quad (28)$$

$$t_p = t_0 \cdot f = 7.188 \times 10^{-48} \cdot 7500 = 5.391 \times 10^{-44} \text{ s} \quad (29)$$

$$(t_p)^2 = 2.906 \times 10^{-87} \text{ s}^2 \quad (30)$$

$$\frac{c^5}{\hbar} = \frac{(2.998 \times 10^8)^5}{1.055 \times 10^{-34}} = 2.297 \times 10^{76} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^4} \quad (31)$$

$$G = 2.906 \times 10^{-87} \cdot 2.297 \times 10^{76} = 6.67430 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \quad (32)$$

(Corrected on 30 Sept. 2026: unit of $c^5/\hbar$ previously $\text{m}^3/(\text{kg} \cdot \text{s}) - [c^5/\hbar] = \text{m}^5 \text{s}^{-5}/(\text{kg} \text{m}^2 \text{s}^{-1}) = \text{m}^3/(\text{kg} \cdot \text{s}^4)$; only then does $\cdot \text{s}^2$ give the unit of G .)

Deviation from the CODATA value: 0.000% (agreement by construction, since $t_0 \cdot f = t_p$ already contains the gravitational constant via the Planck time)

Interpretation: $G \sim t^2$ means: Gravity is linked to the **squared time scale**. This offers an interpretation of why gravity is the weakest force – it is a “slow” process that builds up over long time scales.

Important: $c^5/\hbar$ is **not a physical factor** here, but only the conversion from $[\text{s}^2]$ to $[\text{m}^3/(\text{kg} \cdot \text{s}^2)]$.

Perspective 2: Geometry (Structural Level)

The second perspective derives G from the torsion strain ξ :

$$G = \frac{\xi}{2} \cdot k_{\text{conversion}} \quad (33)$$

Geometric component: $\xi/2$ [dimensionless]

SI conversion: $k_{\text{conversion}}$ [unit conversion]

Derivation from the T0 fundamental formula $\xi = 2\sqrt{G \cdot m}$:

$$\xi^2 = 4Gm \quad (34)$$

$$G = \frac{\xi^2}{4m} = \frac{\xi}{2} \quad \text{with } m = \xi/2 \quad (35)$$

Note on dimensionality: The formula $\xi = 2\sqrt{G \cdot m}$ is formally consistent only in a natural unit system in which the relevant mass m is dimensionless (e.g., in Planck units with $m = \xi/2$ as a dimensionless coupling). In SI units, $2\sqrt{G \cdot m_e}$ has dimensions $\text{m}^{3/2} \text{kg}^{1/2} \text{s}^{-1}$ and therefore cannot be equated directly with the dimensionless ξ . The conversion factor $k_{\text{conversion}}$ carries these units.

Calculation:

$$\xi/2 = \frac{4/30000}{2} = \frac{2}{30000} = 6.667 \times 10^{-5} \text{ (dimensionless)} \quad (36)$$

$$k_{\text{conversion}} = 10^{-6} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \quad (37)$$

$$G \approx 6.667 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \quad (38)$$

Deviation from the CODATA value: -0.11%

(Corrected on 30 Sept. 2026: previously $G \approx 6.674 \times 10^{-11}$ and $0.01\% - 6.667 \times 10^{-5} \cdot 10^{-6} = 6.667 \times 10^{-11}$, i.e. -0.11% relative to CODATA.)

Interpretation: $G \sim \xi$ means: Gravity = lattice deformation. The gravitational strength is directly proportional to the torsion strain of the spacetime lattice. Gravity thus appears as geometry.

Perspective 3: Cosmology (Macro Level)

The third perspective uses a cosmological time scale:

$$G = \frac{k_G}{T \cdot \pi} \quad (39)$$

where:

$$T = 100 \text{ million years} = 3.15576 \times 10^{15} \text{ s} \quad (40)$$

$$k_G = G \cdot T \cdot \pi = 6.617 \times 10^5 \text{ (calculated from Formula 1)} \quad (41)$$

Calculation:

$$G = \frac{6.617 \times 10^5}{3.15576 \times 10^{15} \cdot \pi} = 6.67430 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \quad (42)$$

Deviation from the CODATA value: 0.000% (identical to Formula 1, since k_G is calculated from it)

Interpretation: $G \sim 1/T$ means: Gravity is "diluted" over cosmic time scales. The larger the time scale T , the weaker G appears locally. This connects the micro scale (t_0) with the macro scale (cosmological).

The Equivalence of the Three Formulas

Velocity analogy for illustration:

Consider velocity v :

$$v = \frac{s}{t} \quad (\text{kinematic}) \quad (43)$$

$$v = a \cdot t \quad (\text{dynamic}) \quad (44)$$

$$v = \sqrt{\frac{2E}{m}} \quad (\text{energetic}) \quad (45)$$

All three describe **the same** velocity, just from different perspectives.

Likewise for G :

- **Formula 1 $\equiv$ Formula 3:** Mathematically identical (by definition of k_G)
- **Formula 2 $\approx$ Formula 1:** With conversion factors, $\sim 0.11\%$ difference (*Corrected on 30 Sept. 2026: previously $\sim 0.01\%$ – follow-up value from perspective 2.*)

The role of $\hbar$ and c :

In **all three** formulas, $\hbar$ and c are **only conversion factors** for SI units. The actual physics resides in ξ , f , t_0 , T .

(*Corrected on 30 Sept. 2026: geometry row previously $6.674 - \xi/2 \cdot 10^{-6}$ gives 6.667.*)

The Weak Interaction: W and Z Bosons

The masses of the W and Z bosons are linked to the Higgs mechanism in the Standard Model. In T0 theory, they also have a geometric interpretation.

Perspective	$G [10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2]$	Shows
1. Time structure	6.67430	$G \sim t^2$ (slow)
2. Geometry	6.667	$G \sim \xi$ (deformation)
3. Cosmology	6.67430	$G \sim 1/T$ (diluted)
CODATA (exp)	6.67430	Reference

Table 2: The three perspectives on G – one constant, three viewpoints.

Basic structure:

$$m_W \approx f \cdot \pi^2 \cdot k_W / 1000 \quad (46)$$

$$m_Z \approx f \cdot \pi^2 \cdot k_Z / 1000 \quad (47)$$

The factor $f \cdot \pi^2$ appears because the weak interaction is connected to the surface of the 3-sphere.

Experimental values:

$$m_W = 80.379 \text{ GeV} \quad (48)$$

$$m_Z = 91.1876 \text{ GeV} \quad (49)$$

The ratio:

$$\frac{m_Z}{m_W} = \frac{91.19}{80.38} = 1.134 \quad (50)$$

In the Standard Model, in the on-shell scheme, $m_Z/m_W = 1/\cos\theta_W = 1.134$ holds by definition; the value 1.141 belongs to $\sin^2\theta_W = 0.2312$ in the $\overline{\text{MS}}$ scheme.

The ratio formed above therefore agrees with the Standard Model; the earlier statement of a 0.5% deviation rested on mixing the two schemes.

.6 Stage 3: The Leptons

The Electron: Fundamental Holographic Projection

The electron is the lightest charged lepton. Its mass is $m_e = 0.5109989461 \text{ MeV}$. In T0 theory, it arises as a holographic projection of the Higgs VEV onto the sub-Planck scale.

The fundamental formula:

$$m_e = \frac{v}{f \cdot (2\pi^3 + 3)} \cdot 1000 \quad (51)$$

The factor $2\pi^3 + 3$ describes the three-dimensional nature of the electron:

- $2\pi^3 \approx 62.01$: geometric factor (not a sphere volume) (*Corrected on 30 Sept. 2026: previously "Double volume of a 3D sphere" – the unit ball has volume $4\pi/3 = 4.19$, doubled 8.38; $2\pi^2$ is the surface of the 3-sphere.*)
- +3: Three spatial degrees of freedom

Numerical calculation:

$$2\pi^3 + 3 = 2 \times 31.006 + 3 = 65.012 \quad (52)$$

$$f \cdot (2\pi^3 + 3) = 7491.91 \times 65.012 = 487.08 \times 10^3 \quad (53)$$

$$m_e = \frac{246.71}{487.08 \times 10^3} \cdot 1000 = 0.5065 \text{ MeV} \quad (54)$$

Comparison with experiment: $m_{e,\text{exp}} = 0.5110 \text{ MeV}$

Precision: -0.88% deviation (*Corrected on 30 Sept. 2026: previously $1.02\% - 0.5065/0.51100 - 1 = -0.88\%$. With $f = 7500$ instead of 7491.91 one would get $m_e = 0.5038 \text{ MeV}$, -1.4% .)*

The Muon: Second Generation as Circular Resonance

The muon is approximately 207 times heavier than the electron. In T0 theory, the muon arises as a "circular resonance of second order".

The fundamental formula:

$$m_\mu = v \cdot \frac{\pi}{f} \cdot 1000 \quad (55)$$

Numerical calculation:

$$\frac{\pi}{f} = \frac{3.14159}{7491.91} = 4.194 \times 10^{-4} \quad (56)$$

$$m_\mu = 246.71 \times 4.194 \times 10^{-4} \cdot 1000 = 103.5 \text{ MeV} \quad (57)$$

Comparison with experiment: $m_{\mu,\text{exp}} = 105.66 \text{ MeV}$

Precision: 2.2% deviation (*Note of 30 Sept. 2026: with $f = 7500$ instead of 7491.91 one would get $m_\mu = 102.90 \text{ MeV}$, -2.6% .)*

The Tau: Third Generation

The tau lepton is the heaviest lepton.

The fundamental formula:

$$m_\tau = m_\mu \cdot \left(\frac{4\pi}{3}\right)^2 \quad (58)$$

Numerical calculation:

$$\left(\frac{4\pi}{3}\right)^2 = (4.189)^2 = 17.55 \quad (59)$$

$$m_\tau = 103.5 \times 17.55 = 1816 \text{ MeV} = 1.816 \text{ GeV} \quad (60)$$

Comparison with experiment: $m_{\tau,\text{exp}} = 1.777 \text{ GeV}$

Precision: $+2.2\%$ deviation (*Corrected on 30 Sept. 2026: previously $2.0\% - 1816/1776.9 - 1 = +2.2\%$.)*

Precision through Ratio Reconstruction

A central tool of T0 theory is that **correction values can be back-calculated from mass ratios**, thereby achieving higher accuracy.

The principle:

The T0 formulas normally contain no geometric calibration factors (such as k factors) whose derivation carries uncertainties. However, when we form **ratios** between measured quantities, these factors cancel out.

Worked example 1: Obtaining correction value from empirical lepton masses

T0 theory predicts for lepton masses:

$$m_e = \frac{v}{f \cdot (2\pi^3 + 3)} \cdot k_m \cdot 1000 \quad (61)$$

$$m_\mu = \frac{v \cdot \pi}{f} \cdot k_m \cdot 1000 \quad (62)$$

where k_m is a calibration factor (theoretically $k_m = 1$, but with uncertainty).

Step 1: Back-calculate correction value from electron data

From the experimental electron mass:

$$m_e^{\text{exp}} = 0.5110 \text{ MeV} \quad (63)$$

$$k_m^{\text{rec}} = \frac{m_e^{\text{exp}} \cdot f \cdot (2\pi^3 + 3)}{v \cdot 1000} \quad (64)$$

$$= \frac{0.5110 \cdot 7491.91 \cdot (2\pi^3 + 3)}{246.71 \cdot 1000} \quad (65)$$

$$= \frac{0.5110 \cdot 7491.91 \cdot 65.04}{246.71 \cdot 1000} \quad (66)$$

$$= \frac{249.091}{246.710} = 1.0096 \quad (67)$$

The reconstructed calibration factor is $k_m^{\text{rec}} \approx 1.01$ and thus deviates by about 1% from the theoretical value.

Step 2: Calculate muon with reconstructed factor

With $k_m^{\text{rec}} = 1.0096$:

$$m_\mu^{\text{rec}} = \frac{v \cdot \pi}{f} \cdot k_m^{\text{rec}} \cdot 1000 \quad (68)$$

$$= \frac{246.71 \cdot \pi}{7491.91} \cdot 1.0096 \cdot 1000 \quad (69)$$

$$= 103.5 \cdot 1.0096 = 104.5 \text{ MeV} \quad (70)$$

Comparison:

- With $k_m = 1$: $m_\mu = 103.5 \text{ MeV}$ (deviation: 2.1%)
- With $k_m^{\text{rec}} = 1.0096$: $m_\mu = 104.5 \text{ MeV}$ (deviation: 1.1%)
- Experiment: $m_\mu^{\text{exp}} = 105.66 \text{ MeV}$

The deviation decreases from 2.1% to 1.1%.

Worked example 2: Ratio prediction (k-independent)

Let us form the mass ratio:

$$\frac{m_\mu}{m_e} = \frac{(v \cdot \pi / f) \cdot k_m \cdot 1000}{(v / [f \cdot (2\pi^3 + 3)]) \cdot k_m \cdot 1000} = \frac{\pi \cdot (2\pi^3 + 3)}{1} = \pi \cdot (2\pi^3 + 3) \quad (71)$$

The factor k_m cancels; the ratio is **independent of k_m** :

$$\frac{m_\mu^{\text{theory}}}{m_e} = \pi \cdot (2\pi^3 + 3) = 3.14159 \cdot 65.04 = 204.3 \quad (72)$$

$$\frac{m_\mu^{\text{exp}}}{m_e} = \frac{105.66}{0.511} = 206.8 \quad (73)$$

The deviation of 1.2% comes from geometric approximations, **not** from the calibration factor.

Worked example 3: g-2 reconstruction

The same principle applies to the anomalous magnetic moments. T0 theory predicts:

$$a_e = \frac{S_3 / f}{k_{\text{geom}}} = \frac{4\pi / 7491.91}{k_{\text{geom}}} \quad (74)$$

$$\Delta a_{\mu-e} = \frac{4\pi}{f^{5/3}} \cdot \frac{1}{k_{\text{geom}}} \quad (75)$$

From experimental data:

$$a_e^{\text{exp}} = 1.15965 \times 10^{-3} \quad (76)$$

$$k_{\text{geom}}^{\text{rec}} = \frac{S_3/f}{a_e^{\text{exp}}} = \frac{4\pi/7491.91}{1.15965 \times 10^{-3}} \approx 1.446 \quad (77)$$

(Corrected on 30 Sept. 2026: previously $\approx 2.272 - (4\pi/7491.91)/1.15965 \times 10^{-3} = 1.446$. Conversely, $k_{\text{geom}} = 2\sqrt{2}/\sqrt{\varphi} = 2.2236$ gives $a_e = 7.54 \times 10^{-4}$ (−35 %); the “approximately 2 %” in the abstract and the symmetry-breaking section therefore do not hold.)

Ratio (k-independent):

$$\boxed{\frac{\Delta a_{\tau-\mu}}{\Delta a_{\mu-e}} = \frac{4\pi/f^{4/3}}{4\pi/f^{5/3}} = f^{1/3} = 7491.91^{1/3} = 19.57} \quad (78)$$

The factor k_{geom} cancels. (Note of 30 Sept. 2026: the two individual formulas give $f^{1/3}$ for the ratio (box), but the calculation below uses $f^{1/3} - 1$. With $f^{1/3}$ one would get $a_\tau = 1.16592 \times 10^{-3} + 6.27 \times 10^{-6} \cdot 19.57 = 1.2886 \times 10^{-3}$; the corpus form $f^{1/3} - 1$ (Docs. 018, 150) does not follow from the formulas given here. With $f = 7500$, $f^{1/3} = 19.574$.)

Tau g-2 prediction from ratio:

$$\Delta a_{\mu-e}^{\text{exp}} = (1.16592 - 1.15965) \times 10^{-3} = 6.27 \times 10^{-6} \quad (79)$$

$$\Delta a_{\tau-\mu}^{\text{pred}} = \Delta a_{\mu-e}^{\text{exp}} \times (f^{1/3} - 1) \quad (80)$$

$$= 6.27 \times 10^{-6} \times (19.57 - 1) = 1.164 \times 10^{-4} \quad (81)$$

$$a_\tau^{\text{pred}} = a_\mu^{\text{exp}} + \Delta a_{\tau-\mu}^{\text{pred}} \quad (82)$$

$$= 1.16592 \times 10^{-3} + 1.164 \times 10^{-4} = 1.282 \times 10^{-3} \quad (83)$$

This prediction is **independent of k_{geom}** ; it depends only on the ratio $f^{1/3} - 1$ and the measured values.

Key statement:

- **Absolute predictions** have uncertainties of ~1-2% (from calibration factors)
- **Ratio predictions** are free of these calibration factors (factors cancel)
- **Reconstructed values** reach deviations of 0.1-0.2%

This methodology can be carried over to further T0 predictions: masses, coupling constants, and anomalous moments.

.7 Stage 4: Quarks and Baryons

The Light Quarks: Up and Down

The up and down quarks are the building blocks of protons and neutrons.

Up quark:

$$m_u \approx \frac{f}{4\pi^3} \approx 60.5 \text{ MeV} \quad (84)$$

Down quark:

$$m_d \approx \frac{f}{2\pi^3 \cdot 1.5} \approx 80.6 \text{ MeV} \quad (85)$$

These values lie far above the current quark masses at 2 GeV ($m_u = 2.16 \text{ MeV}$, $m_d = 4.70 \text{ MeV}$): m_u by a factor of 28, m_d by a factor of 17.

(Corrected on 30 Sept. 2026: previously $m_u \approx 2.3 \text{ MeV}$, $m_d \approx 4.8 \text{ MeV}$ and “agree well” – with $f = 7500$ the formulas give 60.5 and 80.6 MeV.)

The Proton and Neutron

The masses of the proton and neutron arise primarily from the energy of quarks and gluons (QCD binding energy), not from the quark masses themselves.

Proton:

$$m_p \approx 938.3 \text{ MeV} \quad (86)$$

Note: The proton mass is dominated by the strong interaction (QCD) and requires complex lattice calculations. A simple geometric formula as for leptons does not exist, since quarks account for only about 1% of the proton mass, while 99% comes from gluon binding energy.

Neutron:

$$m_n \approx m_p + 1.3 \text{ MeV} \approx 939.6 \text{ MeV} \quad (87)$$

The neutron-proton mass difference of approximately 1.3 MeV corresponds to electroweak symmetry breaking and enables beta decay.

.8 Stage 5: The Heavy Quarks

The Strange Quark

$$m_s \approx \frac{f}{(2\pi^2)^2/(5\varphi)} \approx 156 \text{ MeV} \quad (88)$$

Experimental value: $m_{s,\text{exp}} \approx 93 \text{ MeV}$ (at 2 GeV); deviation +67 %.

(Corrected on 30 Sept. 2026: previously $\approx 95 \text{ MeV}$ – with $f = 7500$ the formula gives 155.7 MeV.)

The Charm Quark

$$m_c \approx \frac{f}{\sqrt{2\pi^2}/\varphi} \approx 2.73 \text{ GeV} \quad (89)$$

Experimental value: $m_{c,\text{exp}} \approx 1.27 \text{ GeV}$; deviation +115 %.

(Corrected on 30 Sept. 2026: previously $\approx 1.27 \text{ GeV}$ – with $f = 7500$ the formula gives 2.73 GeV.)

The Bottom Quark

$$m_b \approx \frac{f}{\sqrt{2\pi^2}/\varphi^2} \approx 4.42 \text{ GeV} \quad (90)$$

Experimental value: $m_{b,\text{exp}} \approx 4.18 \text{ GeV}$; deviation +5.7 %.

(Corrected on 30 Sept. 2026: previously $\approx 4.2 \text{ GeV}$ – with $f = 7500$ the formula gives 4.42 GeV.)

The Top Quark: Maximum Coupling

The top quark, with $m_t \approx 173 \text{ GeV}$, is by far the heaviest quark. The T0 formula is simple:

$$m_t = \frac{v}{\sqrt{2}} = \frac{246.71}{1.414} = 174.5 \text{ GeV} \quad (91)$$

Experimental value: $m_{t,\text{exp}} = 172.69 \text{ GeV}$

Precision: 1.0% deviation

(Corrected on 30 Sept. 2026: previously 0.87 % – $174.5/172.69 = 1.010$.)

.9 Stage 6: The Cosmological Constants

[S] This section deals with the cosmic sector. The present corpus deliberately leaves it aside, because the Standard Model does not derive it either (P39); the following statements on dark energy and dark matter are therefore speculative.

Dark Energy as Highest-Order Symmetry Breaking

Dark energy is one of the most puzzling phenomena of modern cosmology. T0 theory proposes the following explanation: dark energy would be the consequence of the **32-fold symmetry breaking** of the torsion crystal.

The fundamental formula:

$$\rho_\Lambda = \frac{\rho_{\text{Planck}}}{f^{32}/\pi^4} \cdot k_\Lambda \quad (92)$$

where $k_\Lambda \approx \pi/2 \approx 1.57$.

The formula predicts: $\rho_\Lambda \approx 7.85 \times 10^{-26} \text{ kg/m}^3$ (Corrected on 1 Oct. 2026: previously $7.96 \times 10^{-27} \text{ kg/m}^3$ – with $\rho_{\text{Planck}} = c^5/(\hbar G^2) = 5.155 \times 10^{96} \text{ kg/m}^3$, $f = 7500$ and $k_\Lambda = \pi/2$ one obtains $5.155 \times 10^{96} \cdot \pi^4/7500^{32} \cdot \pi/2 = 7.85 \times 10^{-26} \text{ kg/m}^3$, with $f = 7491.91$ $8.13 \times 10^{-26} \text{ kg/m}^3$. This is a factor 13 above the measured value, i.e. one order of magnitude off.)

Experimental value: $\rho_{\Lambda,\text{exp}} \approx 5.96 \times 10^{-27} \text{ kg/m}^3$

Given the span of 123 orders of magnitude, the prediction lies in the right order of magnitude.

Dark Matter as Torsion Holding Factor

Instead of new particles, T0 theory postulates a geometric effect for dark matter.

The fundamental formula:

$$H_{\text{DM}} = \frac{\sqrt{f}}{\pi^2/k_{\text{hold}}} \quad (93)$$

For spiral galaxies: $k_{\text{hold}} \approx 2/\pi \approx 0.637$

$$H_{\text{DM}} \approx 5.6 \quad (94)$$

This corresponds approximately to the factor of 5-6 observed in galaxy rotation curves.